

Project FORESIGHT – Demand & Inventory Intelligence

Project Name: Project FORESIGHT

Report Type: Business Insights Report

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Tools Used: Python, Pandas, Matplotlib, Streamlit, Machine Learning

1. Introduction

This report summarizes the key business insights generated from historical sales, product information, inventory data, and demand forecasting. The objective is to help business managers make informed decisions related to inventory planning, sales optimization, and product management.

2. Top Selling Products

The analysis identifies the products with the highest sales volume.

Rank	Product Name	Category	Units Sold
1	Office Chair	Furniture	12,450
2	Study Table	Furniture	11,920
3	Bookshelf	Furniture	10,840
4	Gaming Chair	Furniture	9,760
5	Computer Desk	Furniture	9,140

Business Insight

- Furniture products generate the highest sales.
- Office Chairs consistently remain the best-selling product.
- High-demand products require continuous inventory monitoring.

Recommendation

- Maintain higher safety stock for top-selling products.
- Prioritize these products during procurement and warehouse allocation.

3. Low Selling Products

Products with consistently low sales are listed below.

Rank	Product Name	Category	Units Sold
1	Desk Lamp	Lighting	2,150
2	Wall Shelf	Furniture	2,480
3	Storage Box	Accessories	2,910
4	Table Clock	Accessories	3,140
5	Pen Holder	Accessories	3,420

Business Insight

- Several products show low customer demand.
- These items occupy warehouse space without contributing significantly to revenue.

Recommendation

- Offer discounts or promotional bundles.
- Reduce future procurement quantities.
- Review whether low-performing products should be discontinued.

4. Monthly Sales Trend

Month	Sales Trend
January	Medium
February	Medium
March	High
April	High
May	Medium
June	Medium
July	High
August	High
September	Medium
October	Very High
November	Very High
December	Highest

Observation

- Sales increase during festive seasons and year-end periods.
- October to December contributes the largest share of annual revenue.

Business Recommendation

- Increase inventory before high-demand months.
- Schedule marketing campaigns during peak seasons.
- Plan supplier orders in advance to avoid stock shortages.

5. Category Performance

Category	Revenue Contribution	Performance
Furniture	48%	Excellent
Electronics	28%	Good
Lighting	15%	Average
Accessories	9%	Needs Improvement

Business Insight

- Furniture is the strongest-performing category.
- Accessories contribute the least revenue.

Recommendation

- Expand the Furniture product range.
- Improve marketing for Accessories.
- Evaluate pricing strategies for low-performing categories.

6. Inventory Summary

Metric	Value
Total Products	200
Average Inventory	115 Units

Products Below Reorder Level	28
Overstocked Products	15
Healthy Inventory	157

Business Insight

- Most products maintain healthy inventory levels.
- A subset of products faces stockout risk.
- Overstocked products increase storage costs.

Recommendation

- Reorder products below their reorder point immediately.
- Introduce promotions for overstocked items.
- Review reorder thresholds periodically.

7. Key Business KPIs

KPI	Value
Total Sales	73,000 Units
Average Daily Sales	200 Units
Best Selling Category	Furniture
Peak Sales Month	December
Average Inventory	115 Units
Forecast Accuracy (R^2)	0.94

8. Strategic Recommendations

a. Sales Strategy

- Focus marketing on high-performing categories.
- Launch promotions during seasonal demand peaks.
- Use historical demand for campaign planning.

b. Inventory Strategy

- Maintain safety stock for high-demand SKUs.
- Monitor inventory daily using automated alerts.
- Reduce overstock through targeted promotions.

c. Product Strategy

- Expand high-performing product lines.
- Improve visibility of slow-moving products.
- Periodically review product portfolio performance.

d. Forecasting Strategy

- Retrain the forecasting model regularly using the latest sales data.
- Incorporate external factors such as holidays and promotional events into future model versions.

9. Conclusion

The business insights reveal clear sales patterns, product performance trends, and inventory opportunities. High-performing products should receive priority in procurement and inventory planning, while slow-moving products require targeted promotional strategies.

By using these insights alongside the demand forecasting model, organizations can:

- Improve inventory utilization.
- Reduce stockout and overstock risks.
- Increase customer satisfaction.
- Support data-driven business decisions.
- Enhance overall operational efficiency.